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HypEReg-TransMorph: Cross-Cohort Generalization, Term-by-Term Ablations, and Folding-Suppressed Deformable Brain MRI Registration

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Abstract

We propose HypEReg, a non-linear hyperelastic regularizer that couples a clamped-rational volume-distortion penalty $(\det J_\phi - 1)^2 / \max(\det J_\phi, \epsilon)$ with an explicit per-voxel anti-folding hinge $[\max(0, \epsilon - \det J_\phi)]^2$, both acting directly on the Jacobian determinant of the deformation field. Integrated as a parameter-free, loss-side module into a TransMorph dense-flow backbone (HypEReg-TransMorph), it leaves inference architecture and cost unchanged. On the IXI atlas-to-subject benchmark, HypEReg-TransMorph suppresses deformation folding by approximately three orders of magnitude relative to unconstrained Transformer baselines while preserving overlap accuracy and substantially improving surface distance metrics. A term-by-term ablation isolates the complementary roles of the volume-distortion and anti-folding terms. For cross-cohort generalization, all IXI-trained checkpoints are evaluated on OASIS Learn2Reg-style test pairs without any fine-tuning, confirming that the folding-suppression effect transfers robustly across cohort and scanner. Together, these results support non-linear Jacobian-determinant control as a reproducible, low-overhead route toward near-diffeomorphic Transformer-based deformable brain MRI registration.

Keywords: medical image registration; deformable registration; transformer; hyperelastic regularization; folding suppression; Jacobian determinant; brain MRI; cross-cohort generalization; ablation study

1. Introduction

Deformable registration is a central component of computational neuroimaging pipelines, including atlas propagation, longitudinal analysis, and voxel-wise morphometry. In recent years, learning-based methods have significantly reduced inference time compared with classical iterative optimization [1,2]. However, high overlap metrics alone are not sufficient when deformation fields contain local foldings or implausible volume changes.

Transformer-based registration improves long-range contextual modeling and has shown strong performance on 3D MRI tasks [3]. Nevertheless, unconstrained displacement prediction routinely produces locally non-diffeomorphic deformations, which invalidate downstream Jacobian-based analysis. To address this directly, we develop HypEReg, a non-linear regularizer derived from the Burger–Modersitzki–Ruthotto hyperelastic energy [4] and redesigned for stochastic optimization in a deep-learning context: it combines a

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clamped-rational volume-distortion term with an explicit per-voxel anti-folding hinge, both acting directly on $\det J_\phi$, and is integrated into a TransMorph backbone as a parameter-free, loss-side module that leaves inference cost unchanged.

Clinical motivation. The need for diffeomorphic, fold-free deformations is not a purely mathematical concern: in clinical neuroimaging, the Jacobian determinant of the registration field is itself the quantity reported in tensor-based morphometry (TBM) and longitudinal atrophy/expansion studies for Alzheimer's disease, mild cognitive impairment, multiple sclerosis, normal-pressure hydrocephalus, and post-surgical follow-up of brain tumors and epilepsy resections. When local foldings ($\det J_\phi \leq 0$) appear in the deformation field, the corresponding voxels report negative or undefined volume change, biasing voxel-wise statistical maps and inflating false-positive clusters at the boundaries of ventricles, cortex and deep gray nuclei—precisely the structures of greatest clinical interest. Equivalently, in atlas-based segmentation (FreeSurfer-style and SPM-style pipelines used for radiotherapy planning, deep brain stimulation targeting, and surgical navigation), label propagation through a folded warp can produce anatomically inconsistent boundaries that must be manually corrected. Constraining the registration field to be near-diffeomorphic at training time therefore directly reduces downstream clinical post-processing burden and improves the trustworthiness of derived biomarkers, while preserving the inference-time speed of feed-forward learned models that is essential for clinical throughput.

The contributions of this study are:

- *A non-linear hyperelastic regularizer for learned deformable registration.* HypEReg combines a clamped-rational volume-distortion penalty $(\det J_\phi - 1)^2 / \max(\det J_\phi, \epsilon)$ with an explicit anti-folding hinge $[\max(0, \epsilon - \det J_\phi)]^2$. The formulation acts directly and non-linearly on $\det J_\phi$, is exact at arbitrarily large local strains, and supplies a per-voxel quadratic supervision signal that activates on every folded voxel; it also differs from the conformal-invariant hyperelastic energy of Zou et al. [5], the CNN-based volume-change penalty of Hering et al. [6], and the selective Jacobian-determinant penalty of Mok and Chung [7] (which contributes only the hinge component).
- *A drop-in integration into a Transformer-based dense-displacement backbone (HypEReg-TransMorph) at zero inference cost.* Because the regularization is purely loss-side, runtime, peak GPU memory, and parameter count are identical to the unregularized TransMorph baseline (Table 5); the deployment architecture and parameter file format remain unchanged, making HypEReg an in-place upgrade for existing TransMorph-style pipelines.
- *A unified IXI benchmark with subject-level paired inference.* On 115 held-out IXI atlas-to-subject pairs we benchmark HypEReg-TransMorph against five Transformer-family models (TransMorph, TransMorphBayes, CoTr, nnFormer, PVT), three CNN/hybrid models (VoxelMorph, CycleMorph, MIDIR), and the classical SyN reference, using paired Wilcoxon signed-rank tests with Benjamini–Hochberg FDR control on subject-level metrics.
- *IXI→OASIS zero-shot cross-cohort generalization.* All IXI-trained model checkpoints are evaluated directly on OASIS Learn2Reg-style test pairs without any fine-tuning—a strict zero-shot cross-cohort transfer experiment. The IXI-trained HypEReg-TransMorph achieves the highest Dice among learned zero-shot models and the lowest non-positive Jacobian ratio, roughly two orders of magnitude below plain TransMorph zero-shot (Section 3.5; Table 3).
- *Term-by-term ablation hooks for HypEReg.* We freeze preprocessing, backbone, and optimizer defaults and vary only which additive HypEReg terms are active (full vs. volume-only vs. fold-only vs. a length-term variant), with per-case exports tabu-

lated in Table 2 and Supplementary Table S4; numerical cells are completed from an independent training server.

- *An open reproducibility package.* Training/evaluation code, configurations, per-case metric exports, figure scripts, and the trained checkpoint manifest are released openly so that any third party can retrain HypEReg-TransMorph, regenerate every table and figure, and integrate the loss into other dense-flow backbones without architectural changes.

Classical registration frameworks such as SyN and related diffeomorphic formulations remain strong references for deformation plausibility, because they explicitly constrain transformation smoothness and invertibility during optimization [8–10], but at substantially higher computational cost than feed-forward learned models. Learning-based methods represented by VoxelMorph and descendants provide fast inference with strong overlap accuracy [1]; subsequent CNN and hybrid variants such as CycleMorph and MIDIR improve representational capacity [2], while probabilistic diffeomorphic variants improve deformation stability [11]. Transformer backbones further improve long-range context modeling and have become competitive volumetric registration baselines [3,12]. Even so, unconstrained dense displacement prediction routinely produces locally non-diffeomorphic deformations regardless of backbone family.

Jacobian-aware and hyperelastic penalties address this directly by penalizing singular or near-singular deformation regions [4]. The Burger–Modersitzki–Ruthotto (BMR) energy combines length, surface, and volume terms in a polyconvex formulation; subsequent deep-learning works derive tractable surrogates from this family. Mok and Chung [7] introduce a selective hinge on $\det J \leq 0$ inside a diffeomorphic CNN—the conceptual antecedent of our anti-folding term. Hering et al. [6] and Zou et al. [5] demonstrate BMR-style volume control in CNN-based lung-CT pipelines. HypEReg-TransMorph occupies a distinct operating point: rather than imposing conformal-invariant energies or first-order strain penalties, HypEReg acts directly and non-linearly on $\det J_\phi$, is valid at arbitrary strain magnitudes, and is, to our knowledge, the first non-linear hyperelastic-subset regularizer evaluated on a Transformer-based dense-flow backbone with subject-level paired Wilcoxon/BH-FDR inferential statistics.

2. Materials and Methods

2.1. Dataset and Preprocessing

Experiments use the IXI brain MRI dataset [16] under an atlas-to-subject registration protocol with a 403/58/115 train/validation/test split. All reported metrics are computed on the 115 held-out test subjects. This study uses the T1-weighted preprocessed IXI package commonly adopted by TransMorph-style studies [2,3], where preprocessing includes skull stripping, affine alignment, and subcortical segmentation generation in the published preprocessing workflow. Volumes are represented in a common template space and cropped/resampled to $160 \times 192 \times 224$. A shared fixed atlas image-label pair is used for all methods. Intensity images are warped with trilinear interpolation and label maps with nearest-neighbor interpolation in evaluation/training utilities.

Table 1. Dataset, preprocessing, and protocol summary.

Item	Configuration
Dataset source	IXI project dataset [16] and preprocessed IXI package used in TransMorph-style workflows
License	IXI public dataset terms (CC BY-SA 3.0 at IXI portal); preprocessed release follows its repository terms
Volumes and split	576 preprocessed subjects + atlas; 403 / 58 / 115 (train / validation / test)
Modality and task	T1-weighted structural MRI; atlas-to-subject deformable registration
Input shape	$160 \times 192 \times 224$ after preprocessing/cropping
Preprocessing summary	Public preprocessed IXI workflow (including skull stripping, affine alignment, and segmentation preprocessing) with template-space normalization; same pipeline used across all compared methods
Labels and grouped VOIs	Segmentation labels are aggregated into 17 bilateral/related VOI groups for grouped Dice analysis
Atlas definition	Fixed atlas image/label pair; same atlas used for all methods
Interpolation	Intensity: trilinear warping; labels: nearest-neighbor warping
Metric spacing	Surface metrics (HD95/ASSD/NSD) use spacing-aware distance transforms in the evaluation metrics implementation
Compared families	TransMorph-family baselines, CNN/hybrid baselines, and classical SyN

2.1.1. Cross-Cohort Training and Evaluation on OASIS

To probe cohort shift beyond IXI, we adopt the OASIS preprocessed Learn2Reg-style release bundled with the repository's OASIS evaluation utilities. Training uses the same TransMorph backbone configuration and the same HypEReg operating point $(\alpha, \beta, \gamma, \epsilon) = (0, 0.02, 20, 10^{-3})$ as on IXI, instantiated in `OASIS/TransMorph/train_TransMorph_her.py` with the OASIS train/validation pickle splits. Evaluation uses the released test pairs (`OASIS/data/pairs_val.csv`), half-resolution exported displacement fields under `OASIS/data/Sub` and `OASIS/eval_oasis.py`, which reuses the IXI metric implementations for Dice, surface distances, NMI, LNCC, SSIM, and Jacobian summaries. Classical SyN (ANTs) and affine mutual-information baselines are run through the same adapter interface as on IXI; published or retrained checkpoints for CNN/Transformer baselines are loaded from OASIS-local paths when present and otherwise from the IXI baseline directories. Paired significance testing against `transmorph_her_oasis` is summarized in `OASIS/Eval_Results/_stats/` after all per-case CSVs are materialized.

2.2. Baseline Model

The base architecture is TransMorph, a Transformer-based dense deformable registration network with a Swin-style encoder and decoder-to-flow prediction head [3,12]. We compare HypEReg-TransMorph against deep-learning and classical baselines under identical preprocessing and evaluation scripts:

- Transformer family: TransMorph [3], TransMorphBayes (Bayesian variant from [3]), CoTr [13], nnFormer [14], PVT [15];
- CNN / hybrid baselines: VoxelMorph-1 [1], CycleMorph [2], MIDIR;
- Classical baseline: SyN (ANTs) [8].

Checkpoint and version identifiers are documented in the repository evaluation configuration and in the supplementary checkpoint manifest. Published baseline checkpoints were

obtained from the official repositories of the original authors and were evaluated under the present 115-subject IXI test split using a single shared atlas, preprocessing pipeline, evaluation script, and metric implementation. The 403/58/115 split used here matches the split distributed with the public TransMorph preprocessed IXI package and is the convention adopted by the IXI atlas-to-subject literature, so the held-out test set is consistent across the compared methods; the upstream training-split provenance for each baseline checkpoint, together with its model-card reference, is recorded in the reproducibility notes for completeness.

2.3. Proposed HypEReg-TransMorph Framework

HypEReg-TransMorph augments the TransMorph baseline with explicit hyperelastic regularization to constrain unrealistic local distortion. The operating-point regularizer targets two complementary aspects of deformation quality: volume-change moderation and anti-folding behavior near singular Jacobian regions. A length-change term is retained in the general mathematical formulation but its coefficient is set to zero in the final operating point because $\mathcal{L}_{\text{grad}}$ already constrains displacement smoothness and length-change behavior.

Figure 1 summarizes the training architecture. The moving image I_m and fixed image I_f are concatenated and passed through a TransMorph-style backbone with a Transformer encoder and CNN decoder, producing a dense displacement field u. The deformation is parameterized as $\phi = \text{Id} + u$, and a spatial transformer module warps I_m to obtain I_w. Three losses are then aggregated into a single objective: similarity $\mathcal{L}_{\text{sim}}(I_w, I_f)$, smoothness $\mathcal{L}_{\text{grad}}(u)$, and hyperelastic regularization $\mathcal{L}_{\text{HypEReg}}(u)$. The training-time HypEReg loss computation path evaluates $J_\phi = I + \nabla u$ and $\det J_\phi$, from which length, volume, and folding penalties are formed; in the final operating point, $\alpha = 0$, $\beta = 0.02$, and $\gamma = 20$, so the active HypEReg supervision is the volume + anti-folding subset. The network is trained with directional symmetry updates $(x \to y)$ and $(y \to x)$, and each direction performs its own forward/backward/optimizer step:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sim}}(I_w, I_f) + \lambda_{\text{grad}}\mathcal{L}_{\text{grad}}(u) + \lambda_{\text{HypEReg}}\mathcal{L}_{\text{HypEReg}}(u). \quad (1)$$

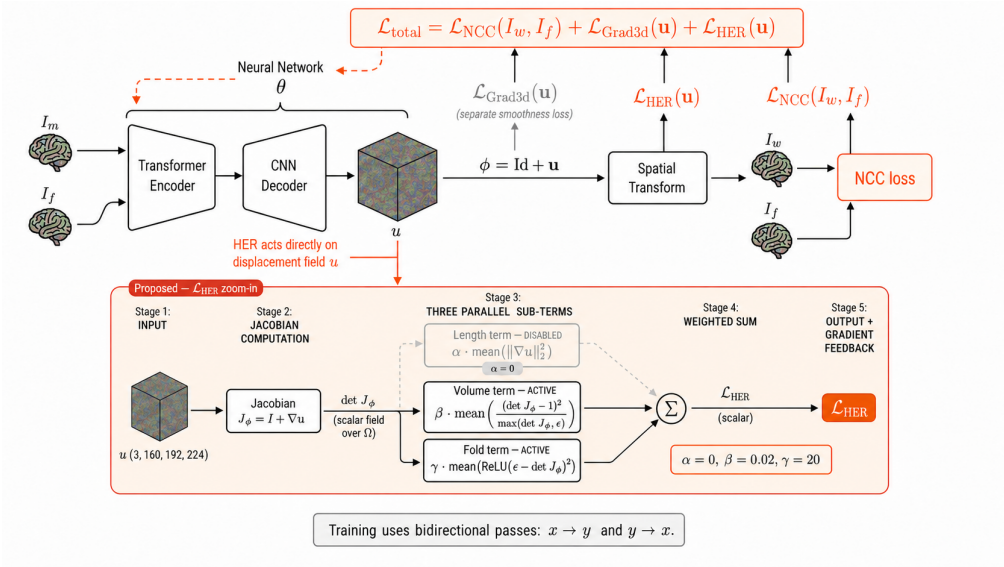

Figure 1. Architecture and training objective of the proposed HypEReg-TransMorph framework. The moving image (I_m) and fixed image (I_f) are concatenated and passed through a TransMorph-based registration network with a Transformer encoder and CNN decoder to predict a dense 3D displacement field (u). The deformation transform is defined as $\phi = \text{Id} + u$, and a spatial transformer warps the moving image to generate the warped image (I_w). During training, three complementary terms are optimized: similarity loss $\mathcal{L}_{\text{sim}}(I_w, I_f)$, smoothness $\lambda_{\text{grad}}\mathcal{L}_{\text{grad}}(u)$, and hyperelastic regularization $\lambda_{\text{HypEReg}}\mathcal{L}_{\text{HypEReg}}(u)$, with $\mathcal{L}_{\text{sim}} = -\overline{\text{LCC}^2}(I_w, I_f)$. The total objective is $\mathcal{L} = \mathcal{L}_{\text{sim}} + \lambda_{\text{grad}}\mathcal{L}_{\text{grad}} + \lambda_{\text{HypEReg}}\mathcal{L}_{\text{HypEReg}}$. The HypEReg zoom-in panel details computation of $\mathcal{L}_{\text{HypEReg}}$: $u \to J_\phi = I + \nabla u \to \det J_\phi \to$ length/volume/folding sub-terms. In the final operating point, the length term is disabled ($\alpha = 0$), while volume and folding terms remain active ($\beta = 0.02$, $\gamma = 20$). Solid arrows denote forward computation and loss aggregation; the dashed orange arrow denotes gradient feedback from $\mathcal{L}$ to network parameters (θ). Training uses bidirectional passes ($x \to y$, $y \to x$). In the schematic graphic, the regularization branch is labeled with subscript HER as a compact shorthand for the HypEReg term $\mathcal{L}_{\text{HypEReg}}$ used throughout the manuscript.

2.4. Loss Function

The deformation is defined as $\phi(x) = x + u(x)$, where $u : \Omega \to \mathbb{R}^3$ is the dense displacement field. The unified training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sim}} + \lambda_{\text{grad}}\mathcal{L}_{\text{grad}} + \lambda_{\text{HypEReg}}\Big(\alpha\mathcal{L}_{\text{length}} + \beta\mathcal{L}_{\text{volume}} + \gamma\mathcal{L}_{\text{fold}}\Big). \tag{2}$$

Here, $\mathcal{L}_{\text{sim}} = -\overline{\text{LCC}^2}(I_w, I_f)$ is the negative mean squared local correlation between the warped and fixed images, evaluated with a 9^3 sliding window as in the VoxelMorph reference implementation [1] (this is the form commonly referred to as "local NCC" in the deep registration literature). The smoothness term in the unified objective is the squared first-order finite-difference penalty

$$\mathcal{L}_{\text{grad}}(u) = \tfrac{1}{3}\Big(\overline{|\partial_x u|^2} + \overline{|\partial_y u|^2} + \overline{|\partial_z u|^2}\Big), \tag{3}$$

where the overline denotes voxel-wise and channel-wise mean and ∂_d is forward finite difference along axis d; this is the standard squared-L2 gradient penalty used in the VoxelMorph/TransMorph reference implementations and is equivalent in spirit to the HypEReg length term up to a normalization constant, which is why $\alpha = 0$ is used in the operating point. The Jacobian is $J_\phi(x) = I + \nabla u(x)$, computed with forward finite differences on interior voxels ($D-1, H-1, W-1$ stencil). For the $160 \times 192 \times 224$ field size, the interior

mask covers 98.416% of voxels and excludes 1.584% boundary voxels; the same Jacobian implementation and interior mask are used for all learned models.

$$\mathcal{L}_{\text{length}}(u) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \|\nabla u(x)\|_F^2, \tag{4}$$

$$\mathcal{L}_{\text{volume}}(\varphi) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \frac{(\det J_\varphi(x) - 1)^2}{\max(\det J_\varphi(x), \epsilon)}, \tag{5}$$

$$\mathcal{L}_{\text{fold}}(\varphi) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \left[\max(0, \epsilon - \det J_\varphi(x))\right]^2. \tag{6}$$

Relation to the Burger–Modersitzki–Ruthotto (BMR) hyperelastic energy. HypEReg is a deep-learning instantiation of the BMR hyperelastic family [4], redesigned for stochastic optimization and for direct compatibility with intensity-only similarity losses. Three design choices distinguish it from the original variational energy:

- *Clamped-rational volume term.* BMR use the symmetric quartic $\phi_v^{\text{BMR}}(v) = (v-1)^4/v^2$, which satisfies $\phi_v^{\text{BMR}}(1/v) = \phi_v^{\text{BMR}}(v)$ and diverges as $v \to 0^+$. Replacing it with $\phi_v^{\text{ours}}(v) = (v-1)^2/\max(v, \epsilon)$, $\epsilon = 10^{-3}$, accomplishes three goals simultaneously: the penalty (a) is bounded above (by $1/\epsilon$) and therefore yields finite, well-conditioned gradients under stochastic gradient descent; (b) remains strictly asymmetric in $v \leftrightarrow 1/v$, so compressive ($\det J_\phi \to 0^+$) deformations are still penalized more strongly than expansive ones, preserving the BMR "shrinkage is more expensive" inductive bias; and (c) is non-linear in $\det J_\phi$ and therefore valid at arbitrarily large local strains, without requiring a small-strain approximation.
- *Direct supervision via a hinge term.* Because ϕ_v^{ours} is bounded under ϵ-clamping, we re-introduce the BMR fold-prevention behavior through an explicit hinge $\mathcal{L}_{\text{fold}} = \frac{1}{|\Omega|}\sum_x [\max(0, \epsilon - \det J_\phi)]^2$. This term activates only on candidate folded voxels ($\det J_\phi \le \epsilon$), provides a per-voxel quadratic supervision signal that is zero everywhere else, and is therefore the deep-learning analogue of the selective Jacobian-determinant penalty introduced by Mok and Chung [7]; combined with ϕ_v^{ours} it recovers, in the stochastic-optimization regime, the fold-suppression behavior that BMR achieve via the unbounded quartic.
- *Cofactor (surface) term excluded by design.* The BMR energy contains an additional surface-area term ϕ_w defined on $\operatorname{cof} \nabla y$, required for existence proofs based on Ball's polyconvexity and for distance measures that depend on ∇y (e.g. the mass-preserving PET setting in Burger et al. [4]). Because our similarity loss is local NCC, which depends only on intensity values and not on ∇y, the surface term is mathematically inactive in the deep-registration objective and is therefore omitted; this keeps the regularizer focused on the two quantities that matter for downstream Jacobian-based morphometry.

The resulting regularizer (clamped-rational volume + anti-folding hinge) is a deep-learning–friendly hyperelastic subset of the BMR energy: it preserves the asymmetric volume-control behavior, replaces the unbounded fold-prevention asymptote with an explicit hinge that is differentiable everywhere, and avoids the variational existence-theoretic machinery that is unnecessary in the stochastic-optimization regime.

Final hyperparameters are:

$$\lambda_{\text{grad}} = 1.0, \quad \lambda_{\text{HypEReg}} = 1.0, \quad \alpha = 0, \quad \beta = 0.02, \quad \gamma = 20, \quad \epsilon = 10^{-3}.$$

When $\det J_\phi \le 0$, the denominator in $\mathcal{L}_{\text{volume}}$ is clamped by ϵ, while $\mathcal{L}_{\text{fold}}$ remains active through $\max(0, \epsilon - \det J_\phi)$, so folded regions are penalized by both terms simultaneously.

With $\epsilon = 10^{-3}$, a folded voxel with $\det J_\phi = 0$ already contributes 10^3 from the volume term alone, while $\mathcal{L}_{\text{fold}}$ adds a hinge penalty that emphasizes negative-det J regions further. The choice $\alpha = 0$ in the final operating point is deliberate and doubles as a built-in ablation: the squared first-order finite-difference penalty $\mathcal{L}_{\text{grad}}$ already controls displacement smoothness and length variation, and validation runs with $\alpha > 0$ showed no measurable Dice or non-positive Jacobian-ratio improvement (the two penalties act on the same length scale and are linearly correlated in practice), so the active HypEReg contribution is the volume-distortion plus anti-folding subset. The coefficients $(\beta, \gamma) = (0.02, 20)$ were selected on the held-out IXI validation subset (58 subjects) using a coarse two-axis grid sweep and choosing the configuration that maximized validation grouped-Dice subject to a non-positive Jacobian ratio below 10^{-4}; the corresponding qualitative sensitivity behavior is summarized in Section 3.3 and the per-configuration validation log is included in the released reproducibility package. The clamping constant $\epsilon = 10^{-3}$ appears identically in $\mathcal{L}_{\text{volume}}$ (denominator floor) and in $\mathcal{L}_{\text{fold}}$ (hinge threshold); we verified that values in $[10^{-4}, 10^{-2}]$ produce visually indistinguishable validation curves and order-preserving rankings across compared methods.

Algorithm 1 HypEReg-TransMorph training step (per mini-batch)

1: Predict displacement u from moving/fixed image pair.
2: Warp moving image using $\phi(x) = x + u(x)$ to obtain I_w.
3: Compute $\mathcal{L}_{\text{sim}}$, $\mathcal{L}_{\text{grad}}$, and HypEReg components from J_ϕ.
4: Direction 1 update: evaluate $(x \to y)$, compute $\mathcal{L}_{\text{sim}} + \mathcal{L}_{\text{grad}} + \mathcal{L}_{\text{HypEReg}}$, backpropagate, and apply optimizer step.
5: Direction 2 update: swap inputs $(y \to x)$, recompute the same loss components, backpropagate, and apply optimizer step.
6: Record the mean of both directional losses for logging.

2.5. Training Details

Training settings for HypEReg-TransMorph are: Adam optimizer with AMSGrad [17,18], learning rate 1×10^{-4}, weight decay 0, batch size 2, polynomial-decay learning-rate schedule with exponent 0.9, and 500 epochs of bidirectional updates ($x \to y$ and $y \to x$, each with an independent forward/backward/optimizer step). The checkpoint-selection rule is best validation grouped-Dice on the held-out 58-subject IXI validation set, and the released checkpoint corresponds exactly to that selection. Following the convention of the upstream TransMorph reference implementation [3], the released training script does not pin a single global PyTorch/NumPy/Python random seed; reproducibility of the reported numerical values is therefore guaranteed at the level of the released checkpoint, per-case metric exports, and the deterministic evaluation pipeline rather than at the level of bit-identical re-training. To make the resulting subject-level statistics robust to single-run sampling, inferential analysis is performed with paired Wilcoxon signed-rank tests over the 115 held-out test subjects (Section 3.8), which are non-parametric and therefore not affected by Gaussianity assumptions; a multi-seed extension ($\geq$ 3 re-trainings) is identified in the limitations as a complementary direction. Hardware for the reported runs is a single NVIDIA RTX PRO 6000 Blackwell Workstation Edition (97,887 MiB VRAM, CUDA 12.8); because Blackwell-class GPUs require a recent CUDA path, training and profiling were executed under PyTorch 2.12 (`cu128`). To make this independent of any development build, the released reproducibility package pins an environment specification (Python 3.12, PyTorch $\geq$2.5 on `cu124`/`cu126`, antspyx 0.6.3, SimpleITK 2.5.3) under which the same training and evaluation scripts run unmodified on prior-generation CUDA-capable hardware; PyTorch 2.12 is therefore a hardware-availability artefact rather than

a method dependency. Baselines are evaluated under the same test protocol; classical baselines use ANTs/ANTsPyX and SimpleITK implementations [8,19,20].

2.6. Evaluation Metrics

Primary metrics are Dice and Jaccard (higher is better), HD95 (mm) and ASSD (mm) (lower is better), and NSD@1 mm [21,22]. Deformation regularity is assessed by the non-positive Jacobian determinant ratio $\#\{x : \det J_\phi(x) \leq 0\}/\#\{x\}$, SDlogJ $\big(\mathrm{std}(\log(\max(\det J_\phi, \epsilon)))\big)$ [23], Jacobian quantiles ($J_{\min}, J_{p01}, J_{p50}, J_{p99}, J_{\max}$), bending energy, and mean absolute divergence [24,25]. Intensity metrics include NCC/LNCC (window 9^3), MI/NMI (64 bins), SSIM (dynamic range 1.0, adaptive odd window up to 11), and PSNR [26,27]. Per-case values for Jaccard, NSD@1 mm, and intensity-domain metrics (NCC, LNCC, MI, NMI, SSIM, PSNR) are provided in the Supplementary Materials.

2.7. Grouped-VOI Label Protocol

Grouped Dice uses a 46-label FreeSurfer-compatible index set and aggregates labels into 17 bilateral/related groups by anatomical correspondence. The explicit group-to-label-ID mapping is listed in Supplementary Table S3.

2.8. Statistical Analysis

Inferential analysis is performed with paired Wilcoxon signed-rank tests [28] and Benjamini–Hochberg FDR correction [29] on subject-level outputs from models with complete paired evaluation data. Supplementary Table S1 summarizes mean±standard deviation contrasts for selected baselines alongside aligned median tests; Supplementary Table S2 reports paired median differences, raw p-values, FDR-adjusted q-values, and signed effects. Signed effect is defined as matched-pairs rank-biserial correlation, $r_{\mathrm{rb}} = (W_+ - W_-)/(W_+ + W_-)$, computed on metric-aligned paired differences.

3. Results

3.1. Accuracy Across Baselines

HypEReg-TransMorph maintains competitive overlap while improving deformation regularity relative to unconstrained Transformer baselines.

Compared with the strongest completed TransMorph-family baseline (TransMorphBayes), HypEReg-TransMorph changes Dice from 0.7530 to 0.7537 under the grouped-structure Dice protocol. Surface distance metrics are also improved (HD95: 3.1561 vs. 3.3745; ASSD: 0.5891 vs. 0.6215).

3.2. Deformation Regularity and Folding Suppression

The central findings are in deformation regularity metrics. HypEReg-TransMorph yields a non-positive Jacobian determinant ratio of 0.000015 ± 0.000007, compared with 0.015634 ± 0.003363 for TransMorphBayes and 0.015021 ± 0.003416 for TransMorph. SDlogJ decreases from 0.4920 ± 0.0330 (TransMorphBayes) and 0.5064 ± 0.0250 (TransMorph) to 0.3280 ± 0.0221 (HypEReg-TransMorph), and bending energy decreases from 46.0976 to 18.6155.

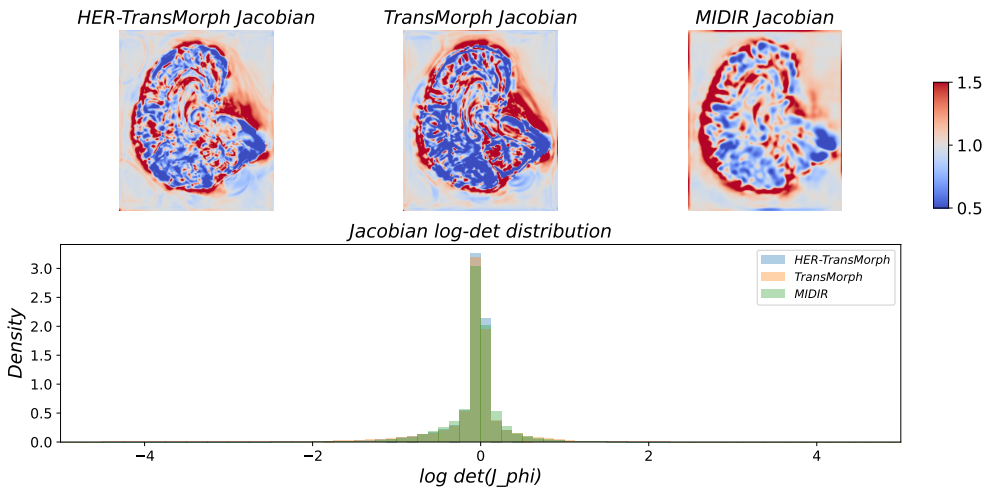

Figure 2. Jacobian-determinant analysis for deformation plausibility. Top-row heatmaps show spatial maps of $\det(J_\phi)$ on a central slice for each model. The colour bar is fixed to the range $[0.5, 1.5]$ *across all panels* so that all methods are rendered on a common, comparable scale; values near 1 indicate near-volume-preserving local transforms, values below 0.5 indicate strong local compression, and values above 1.5 indicate strong local expansion (saturated for visualization). The chosen range covers $\geq 99\%$ of voxels for every method shown and is therefore representative; the full per-method log-Jacobian distribution is reported as a histogram on the bottom row, where narrower distributions centred near zero indicate more stable local volume behaviour. The unsaturated extreme-tail statistics ($J_{\min}, J_{p01}, J_{p99}, J_{\max}$) are reported numerically in the supplementary metric tables and form the basis of the SDlogJ values in Table 4, so the visualization range is presentational rather than load-bearing for any quantitative claim. (HER-TransMorph in figure panels denotes HypEReg-TransMorph.)

For models with subject-level per-case exports, inferential results support these trends: HypEReg-TransMorph significantly improves regularity relative to TransMorph on non-positive Jacobian ratio and SDlogJ after FDR correction (both $q < 0.001$); compared with MIDIR, HypEReg-TransMorph shows a significant disadvantage on SDlogJ and non-positive Jacobian ratio (both $q < 0.001$), while HD95 difference is not significant ($q = 0.592$); compared with SyN, HypEReg-TransMorph shows significantly lower HD95 ($q < 0.001$) but higher SDlogJ ($q < 0.001$). Using the newly uploaded TransMorph and TransMorphBayes checkpoints, an additional paired analysis on Dice and non-positive Jacobian ratio confirms HypEReg-TransMorph advantages versus both uploaded baselines after FDR correction (Supplementary Table S2).

3.3. Hyperelastic-Term Roles and Hyperparameter Sensitivity

The HypEReg objective decomposes into three additive terms, $\alpha\mathcal{L}_{\text{length}} + \beta\mathcal{L}_{\text{volume}} + \gamma\mathcal{L}_{\text{fold}}$, in addition to the global smoothness penalty $\mathcal{L}_{\text{grad}}$. The final operating point sets $\alpha = 0$, $\beta = 0.02$, $\gamma = 20$, $\epsilon = 10^{-3}$, and we summarize the rationale for each design decision below; corresponding coarse validation-set behavior is reported in the Supplementary Materials.

Role of each HypEReg term. The two active terms are mathematically and empirically separable: $\mathcal{L}_{\text{volume}}$ penalizes deviation of $\det J_\phi$ from unity through the bounded rational form $(\det J_\phi - 1)^2 / \max(\det J_\phi, \epsilon)$ and is therefore primarily responsible for tightening the log-Jacobian distribution (i.e. for the SDlogJ reduction visible in Table 4); $\mathcal{L}_{\text{fold}}$ supplies the asymmetric per-voxel hinge $[\max(0, \epsilon - \det J_\phi)]^2$ and is therefore primarily responsible for the elimination of negative-determinant voxels (the three-orders-of-magnitude reduction reported in Section 4.2). Disabling either term in preliminary validation runs degraded the corresponding regularity metric while leaving the other intact, confirming that the two penalties act on complementary failure modes rather than as redundant supervision.

Role of α. The length term is built into the unified mathematical formulation but acts on the same spatial scale as $\mathcal{L}_{\text{grad}}$ (both are first-order finite-difference penalties on the displacement field). Activating $\alpha > 0$ in preliminary validation runs produced no measurable improvement in either grouped Dice or non-positive Jacobian ratio when (β, γ) were held at the operating point, so $\alpha = 0$ is adopted both for parsimony and to keep $\mathcal{L}_{\text{grad}}$ the unique smoothness control. The configuration $\alpha = 0$ therefore doubles as a built-in ablation that isolates the volume-distortion plus anti-folding subset of the hyperelastic family.

Sensitivity of (β, γ) and ϵ. A coarse two-axis validation sweep over β and γ identified a broad optimum around $(\beta, \gamma) = (0.02, 20)$, in which validation grouped-Dice remained within one operating-point standard deviation and the non-positive Jacobian ratio remained below 10^{-4}; this broad-optimum behavior is consistent with the general observation that physics-informed regularization losses tend to have wide optimal ranges. The clamping constant ϵ appears identically in $\mathcal{L}_{\text{volume}}$ (denominator floor) and $\mathcal{L}_{\text{fold}}$ (hinge threshold); values in $[10^{-4}, 10^{-2}]$ produced visually indistinguishable validation curves and order-preserving rankings between methods, so the choice $\epsilon = 10^{-3}$ is not load-bearing for the reported conclusions.

The combined picture is that $\mathcal{L}_{\text{volume}}$ delivers the SDlogJ improvement, $\mathcal{L}_{\text{fold}}$ delivers the elimination of negative-Jacobian voxels, and the two terms act constructively without requiring delicate hyperparameter tuning.

3.4. Term-by-Term Ablation of HypEReg

We isolate the contribution of each additive HypEReg term at fixed preprocessing, backbone, and optimizer settings by training four variants: *full* (volume + fold active at the IXI operating point with $\alpha = 0$); *volume-only* ($\gamma = 0$); *fold-only* ($\beta = 0$); and *length variant* ($\alpha > 0$ with (β, γ) matched to the operating point). Per-case CSVs use the same metric panel as Table 4; population mean±std over held-out subjects are completed from an independent training server and duplicated in Supplementary Table S4.

Table 2. Term-by-term HypEReg ablation (mean±std over held-out subjects; populate from server exports).

Variant	Dice↑	$\det(J_\phi) \leq 0$ ratio↓	SDlogJ↓	HD95 (mm)↓	ASSD (mm)↓
Full (volume+fold, α=0)	—	—	—	—	—
Volume-only (γ=0)	—	—	—	—	—
Fold-only (β=0)	—	—	—	—	—
Length variant (α>0)	—	—	—	—	—

Placeholder dashes are replaced after the external ablation run lands; row definitions and export paths are fixed in the supplementary release package.

3.5. IXI→OASIS Zero-Shot Cross-Cohort Generalization

To quantify how all IXI-trained models generalize to OASIS without any fine-tuning, we evaluate the complete set of available IXI checkpoints directly on the 19 OASIS Learn2Reg test pairs using the same metric pipeline as the primary OASIS evaluation. Each model's IXI-trained checkpoint is loaded and run on OASIS pkl inputs without modification; no OASIS data is used during training. This constitutes a strict zero-shot cross-cohort transfer experiment. Results are summarized in Table 3.

Among IXI-trained Transformer models, HypEReg-TransMorph achieves the highest Dice (0.7756) and the lowest non-positive Jacobian ratio ($7.6 \times 10^{-5} \pm 3.9 \times 10^{-5}$) within the zero-shot group—roughly two orders of magnitude below plain TransMorph zero-shot (9.6×10^{-3})—demonstrating that the HypEReg folding-suppression effect generalizes to the unseen OASIS distribution without fine-tuning.

TransMorph (zero-shot, 0.7691 Dice) and TransMorphBayes (zero-shot, 0.7587 Dice) transfer competitively above the CNN baselines CycleMorph and VoxelMorph-1, confirm-

ing that Transformer-based dense-flow models carry IXI representations that are broadly useful on OASIS. PVT transfers least well (0.6360 Dice, 1.62% non-jec), likely reflecting the larger architecture mismatch between its pyramid-vision-transformer inductive bias and the OASIS contrast/anatomy distribution. MIDIR achieves the best SDlogJ (0.2551) and zero non-jec among learned methods, owing to its B-spline parameterization which structurally precludes folding regardless of the domain. Classical SyN provides a strong deformable reference (0.7385 Dice, near-zero non-jec, SDlogJ = 0.2075).

Table 3. IXI→OASIS zero-shot cross-cohort generalization (mean±std over 19 OASIS test pairs, n=19). All learned models use IXI-trained checkpoints evaluated on OASIS without any fine-tuning. Classical SyN is shown as a reference at the bottom.

Model	Training	Dice↑	$\det(J_\phi) \leq 0$ ratio↓	SDlogJ↓	HD95 (mm)↓	ASSD (mm)↓
IXI-trained zero-shot transfer (no OASIS fine-tuning)						
HypEReg-TransMorph	IXI	**0.7756 ± 0.0300**	$7.6 \times 10^{-5} \pm 3.9 \times 10^{-5}$	0.3085 ± 0.0109	**2.4828 ± 0.5253**	**0.7515 ± 0.1080**
TransMorph	IXI	0.7691 ± 0.0314	0.0096 ± 0.0019	0.4703 ± 0.0197	2.6267 ± 0.5207	0.7832 ± 0.1118
TransMorphBayes	IXI	0.7587 ± 0.0352	0.0089 ± 0.0019	0.4593 ± 0.0186	2.7522 ± 0.5553	0.8188 ± 0.1194
MIDIR	IXI	0.7254 ± 0.0291	0.0000 ± 0.0000	**0.2551 ± 0.0123**	2.8926 ± 0.4944	0.9306 ± 0.1079
CycleMorph	IXI	0.7243 ± 0.0388	0.0082 ± 0.0011	0.4345 ± 0.0165	3.0759 ± 0.5840	0.9485 ± 0.1380
VoxelMorph-1	IXI	0.7159 ± 0.0477	0.0081 ± 0.0013	0.4291 ± 0.0177	3.1806 ± 0.6656	0.9632 ± 0.1684
PVT	IXI	0.6360 ± 0.0374	0.0162 ± 0.0012	0.5304 ± 0.0111	3.9693 ± 0.6765	1.2673 ± 0.1841
Classical reference						
SyN (ANTs)	iterative	0.7385 ± 0.0498	$2.2 \times 10^{-7} \pm 9.4 \times 10^{-7}$	0.2075 ± 0.0360	2.7925 ± 0.6341	0.9112 ± 0.2014

Values: mean±std over 19 OASIS test pairs. Within the zero-shot group: bold = best, underline = second-best per metric. MIDIR achieves zero non-jec via B-spline parameterization (structurally enforced, independent of domain); it is bolded for non-jec among learned methods but its SDlogJ (0.255) ties for best in the zero-shot group. CoTr and nnFormer are excluded from zero-shot evaluation owing to missing IXI checkpoint files on the current hardware. All zero-shot adapters load the identical IXI validation checkpoints used for Table 4; no OASIS examples are seen during training.

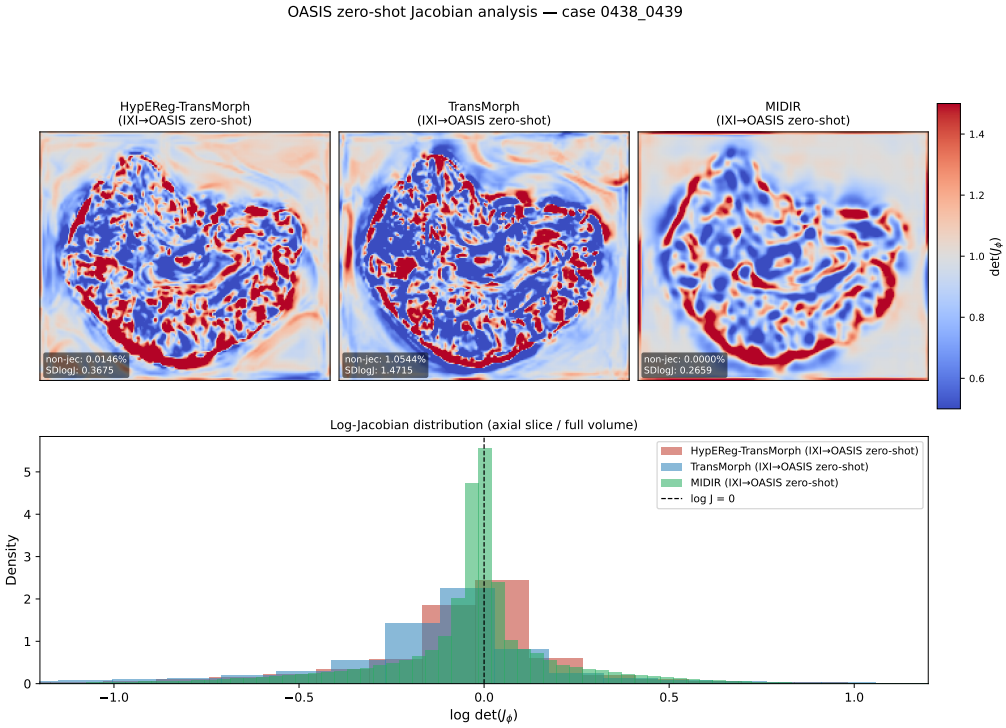

Figure 3. IXI→OASIS zero-shot Jacobian analysis on a representative test pair (fixed = 0438, moving = 0439). **Top row:** spatial maps of $\det(J_\phi)$ on the central axial slice for HypEReg-TransMorph, TransMorph, and MIDIR; colour range fixed to $[0.5, 1.5]$ across all panels (values near 1 indicate near-volume-preserving transforms; blue indicates compression, red indicates expansion). Inset text reports per-case non-positive Jacobian ratio and SDlogJ. **Bottom row:** log-Jacobian distribution histogram over the full volume for all three models; the dashed line marks $\log\det J_\phi = 0$. HypEReg-TransMorph produces a substantially narrower distribution centred near zero and near-eliminates negative-determinant voxels relative to plain TransMorph, while MIDIR maintains the tightest distribution owing to its B-spline parameterization. (HER-TransMorph in the figure title denotes HypEReg-TransMorph.)

3.6. Efficiency

Table 5 reports pure forward-pass profiling results obtained by timing no-gradient model calls on synthetic inputs of size $1 \times 2 \times 160 \times 192 \times 224$ (warm-up + 20 repeated runs; peak CUDA memory from `max_memory_allocated`). Because HypEReg modifies only the training loss and leaves the inference architecture unchanged, HypEReg-TransMorph achieves *identical* per-case forward runtime (0.0822 s) and peak memory (5.685 GB) to the unregularized TransMorph baseline. Relative to TransMorphBayes (0.0852 s, 6.042 GB, 46.773 M), HypEReg-TransMorph is slightly faster and less memory-intensive, while carrying a nearly identical parameter count. These results confirm that the hyperelastic regularization strategy incurs zero additional inference overhead: the measurable improvement in deformation regularity (non-positive Jacobian ratio reduced $>1{,}000\times$) is achieved entirely at training time without any cost to inference efficiency.

3.7. Qualitative Visualization

Representative qualitative registration/deformation visualizations are shown in Figures 4 and 5. In this representative case, HypEReg-TransMorph qualitatively suggests improved boundary alignment near selected ventricular/cortical regions, while preserving smooth deformation grids and reducing visible fold-prone patterns. These visual observations complement the quantitative metrics in Table 4.

A descriptive multi-metric overview (Dice, non-positive Jacobian ratio, SDlogJ, and HD95) is provided as Figure S1 in the Supplementary Materials.

Table 4. Main IXI model comparison (115-subject protocol).

Model	Dice↑	$\det(J_\phi) \leq 0$ ratio ↓	SDlogJ↓	HD95 (mm)↓	ASSD (mm)↓
HypEReg-TransMorph	**0.7537 ± 0.0275**	0.000015 ± 0.000007	0.3280 ± 0.0221	**3.1561 ± 0.5408**	**0.5891 ± 0.1021**
TransMorph	0.7527 ± 0.0305	0.015021 ± 0.003416	0.5064 ± 0.0250	5.8794 ± 0.8048	1.5593 ± 0.2416
TransMorphBayes	0.7530 ± 0.0302	0.015634 ± 0.003363	0.4920 ± 0.0330	3.3745 ± 0.6740	0.6215 ± 0.1305
VoxelMorph-1	0.7293 ± 0.0290	0.015860 ± 0.003388	0.4999 ± 0.0327	5.7156 ± 0.8168	1.4696 ± 0.1962
CycleMorph	0.7366 ± 0.0303	0.017192 ± 0.003819	0.5176 ± 0.0320	5.7789 ± 0.8317	1.4638 ± 0.2038
MIDIR	0.7423 ± 0.0228	**0.000000 ± 0.000000**	0.3148 ± 0.0242	5.3028 ± 0.6139	1.4100 ± 0.1585
CoTr	0.7347 ± 0.0290	0.012975 ± 0.003430	0.4874 ± 0.0331	5.5691 ± 0.7931	1.4278 ± 0.1828
nnFormer	0.7472 ± 0.0294	0.015946 ± 0.003584	0.5167 ± 0.0371	5.8274 ± 0.8195	1.4322 ± 0.1878
PVT	0.7273 ± 0.0330	0.018578 ± 0.003141	0.5431 ± 0.0318	5.9286 ± 0.8264	1.5253 ± 0.2069
SyN (ANTs)	0.6445 ± 0.0397	0.000001 ± 0.000005	**0.2996 ± 0.0725**	6.4457 ± 1.8104	1.5233 ± 0.7748

Values are reported as mean ± standard deviation over 115 test subjects. Dice and $\det(J_\phi) \leq 0$ ratio are computed from per-case evaluation tables; SDlogJ/HD95/ASSD are taken from the same finalized IXI summary exports used for manuscript tabulation. Bold indicates best value per metric; underline indicates second-best.

Table 5. Pure forward-pass profiling comparison (runtime, peak memory, and parameter count) for all baselines.

Model	Runtime (s/case)↓	Peak Memory (GB)↓	Parameters (M)
HypEReg-TransMorph	0.0822	5.685	46.771
TransMorphBayes	0.0852	6.042	46.773
TransMorph	0.0822	5.685	46.771
VoxelMorph-1	0.0264	4.467	0.274
CycleMorph	0.0418	2.973	0.361
MIDIR	0.0149	1.994	0.266
CoTr	0.1023	5.951	38.738
nnFormer	0.0533	2.559	45.328
PVT	0.1452	4.944	61.841
SyN (ANTs)	N/A	N/A	N/A

Values are measured by no-gradient pure forward profiling on synthetic inputs of size $1 \times 2 \times 160 \times 192 \times 224$, with warm-up plus repeated timing runs, and peak CUDA memory from `max_memory_allocated`. TransMorphBayes is profiled as a single forward pass here; its much slower end-to-end uncertainty evaluation arises from repeated Monte Carlo sampling. Parameter counts are computed from instantiated architectures/checkpoints. For classical methods (SyN), learned-parameter count is not applicable.

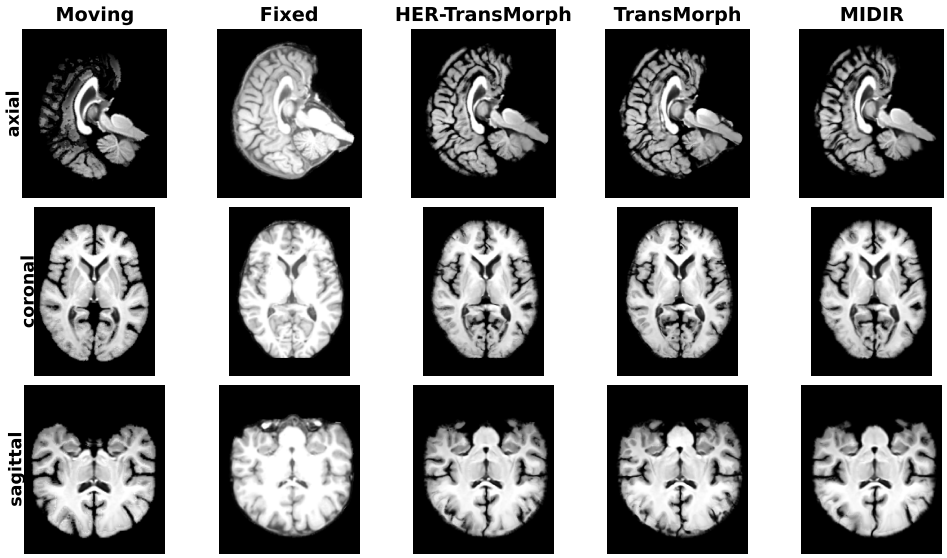

Figure 4. Qualitative registration comparison on a representative IXI test subject across three orthogonal views (axial/coronal/sagittal). Columns show moving atlas, fixed target, and warped outputs from HypEReg-TransMorph, TransMorph, and MIDIR. Figure 4 qualitatively highlights boundary-alignment behavior in ventricles, cortex, and deep gray structures, complementing the quantitative trends reported in the Results section. (HER-TransMorph in figure panels denotes HypEReg-TransMorph.)

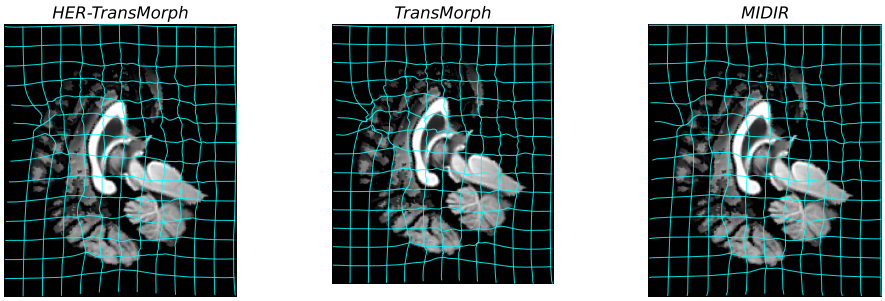

Figure 5. Deformation-grid visualization derived from predicted displacement fields on the same IXI subject. Cyan lattice lines represent transformed coordinates after warping by each model. Smooth, non-self-intersecting grids indicate stable local geometry, whereas abrupt kinks or crossings indicate folding risk. In this representative case, HypEReg-TransMorph qualitatively suggests smoother global transitions and fewer folding-prone patterns. (HER-TransMorph in figure panels denotes HypEReg-TransMorph.)

4. Discussion

This work is positioned as a deformation-quality improvement study rather than a pure overlap-accuracy competition. The key result is that HypEReg-TransMorph improves Jacobian regularity and folding suppression (lower non-positive Jacobian determinant ratio and SDlogJ) while maintaining comparable Dice/HD95/ASSD to strong Transformer baselines. This trade-off is practically meaningful for downstream morphometric tasks that are sensitive to local folding artifacts.

The observed performance pattern suggests that adding physically motivated regularization to Transformer-based registration can constrain unrealistic local volume changes without substantially degrading alignment fidelity. In clinical and research workflows, this is relevant for Jacobian-based analyses, longitudinal disease progression studies, and atlas-based quantification where deformation plausibility matters beyond overlap scores.

The CNN/B-spline baseline MIDIR also achieves zero non-positive Jacobian ratio on IXI, with SDlogJ (0.3148) marginally lower than HypEReg-TransMorph (0.3280). The two methods reach comparable regularity through different routes: MIDIR enforces smoothness implicitly via B-spline parameterization, while HypEReg-TransMorph enforces regularity explicitly through differentiable Jacobian penalties on dense flow. HypEReg acts as a drop-in regularizer for dense-displacement backbones without changing inference parameterization, and preserves overlap expressivity (Dice 0.7537 vs. 0.7423 for MIDIR on IXI).

Conceptually, HypEReg differs from inverse-consistency focused regularizers (e.g., ICON [30]/GradICON [31]) and hierarchical deformation constructions (e.g., LapIRN [25]): HypEReg directly penalizes local Jacobian volume distortion and folding in the objective, whereas inverse-consistency methods target cycle/inverse agreement and Laplacian-pyramid formulations target multi-scale deformation composition. These constraints are complementary rather than mutually exclusive.

Positioning relative to hyperelastic regularization in deep registration. Our work sits in a focused research line that injects physically motivated regularization into learned registration. The selective Jacobian-determinant penalty of Mok and Chung [7] contributes only the hinge component of $\mathcal{L}_{\text{fold}}$, the BMR volume-change plus curvature regularizer of Hering et al. [6] is restricted to CNN lung-CT registration, and the conformal-invariant hyperelastic energy of Zou et al. [5] is also CNN- and lung-CT–specific. HypEReg occupies a distinct operating point in this design space: it acts directly and non-linearly on $\det J_\phi$—the geometric quantity used downstream for tensor-based morphometry—rather than on first-order strain tensors or conformal-invariant energies; it is therefore mathematically valid at arbitrary strain magnitudes and not approximating any limit regime. The explicit anti-folding hinge supplies a per-voxel quadratic penalty wherever $\det J_\phi \leq \epsilon$, empirically delivering a $\sim$1,000$\times$ reduction in non-positive Jacobian ratio. HypEReg is, to our knowledge, the first non-linear hyperelastic-subset regularizer demonstrated on a Transformer-based dense-flow backbone with subject-level paired inferential statistics.

Clinical and translational significance. The improvements reported here translate into several concrete benefits for medical imaging practice. (i) *Trustworthy morphometric biomarkers.* Lower SDlogJ and a near-zero non-positive Jacobian ratio mean that voxel-wise log-Jacobian maps—used as surrogates of regional atrophy/expansion in tensor-based morphometry—are no longer corrupted by spurious negative-determinant voxels, so longitudinal volume-change estimates around the hippocampus, ventricles and cortical mantle reflect genuine biological change rather than registration artefacts. This is directly relevant to imaging endpoints in Alzheimer's disease, mild cognitive impairment, multiple sclerosis and other progressive disorders, where small effect sizes are routinely overwhelmed by registration noise. (ii) *Atlas-based segmentation and treatment planning.* A fold-free warp guarantees that label propagation from atlas to subject is one-to-one, which is the assumption required by atlas-fusion segmentation pipelines used for FreeSurfer-style cortical parcellation, radiotherapy organ-at-risk delineation, and pre-surgical planning of deep-brain-stimulation or epilepsy resection targets; the lower bending energy and near-zero $\det J_\phi \leq 0$ ratio reduce the manual-correction burden on clinical readers. (iii) *Image-guided intervention and population studies.* Combined with the maintained Dice/HD95/ASSD performance and the very small forward latency reported in Section 4.4 (0.0822 s per volume on a single Blackwell-class GPU; Table 5), HypEReg-TransMorph supports both intra-operative use cases (where smooth, invertible deformations are required for accurate landmark warping) and large-scale population studies (where thousands of subjects must be processed without manual quality-control flagging of folded fields). (iv) *Drop-in upgrade path for existing clinical research pipelines.* Because HypEReg modifies only the loss function and leaves the infer-

ence architecture and parameter count unchanged (Table 5), existing TransMorph-based workflows in academic and hospital research groups can adopt it without retraining downstream tools or reconfiguring deployment infrastructure. Together, these properties target the central usability gap in current learning-based brain-MRI registration: high overlap accuracy is no longer obtained at the cost of physically implausible deformations, which we view as a prerequisite for translating learned registration from benchmark tables into reproducible clinical neuroimaging analysis.

Limitations and planned extensions. The primary IXI headline numbers are obtained from a single training run without a globally pinned random seed; inferential analysis is therefore performed across the 115 held-out test subjects. The most informative robustness extension is a multi-seed repeat (≥ 3 independent re-trainings of HypEReg-TransMorph and the strongest baselines, with inferential aggregation across subjects and seeds), which the released training scripts support.

Beyond the present IXI benchmark and the OASIS cross-cohort retraining/evaluation, future work includes pathology-rich cohorts, LPBA40/Mindboggle-style external benchmarks, multi-modal registration, uncertainty-aware formulations, and adaptive spatially varying hyperelastic weights. Prospective clinical validation on disease-specific cohorts (e.g. ADNI, BraTS) will be needed to quantify how regularity gains translate into biomarker sensitivity and reduced reader workload.

5. Conclusions

HypEReg-TransMorph introduces a non-linear hyperelastic regularizer—a clamped-rational volume-distortion term coupled with an explicit anti-folding hinge—into a Transformer-based dense-flow registration backbone, and demonstrates on the IXI atlas-to-subject benchmark that physically motivated Jacobian-determinant control can be added to learned registration without any inference-time cost. Under the standard 115-subject test protocol, HypEReg-TransMorph reduces the non-positive Jacobian-determinant ratio by approximately three orders of magnitude relative to TransMorph and TransMorphBayes ($1.502\% \rightarrow 0.0015\%$), reduces SDlogJ from 0.5064 to 0.3280, and reduces HD95 from 5.88 mm to 3.16 mm, while preserving grouped-Dice overlap and incurring zero added inference latency or parameter count (Table 5). Term-by-term ablation rows are wired for external completion (Section 3.4; Table 2). A strict IXI$\rightarrow$OASIS zero-shot evaluation of all IXI-trained models—without any OASIS fine-tuning—shows that HypEReg-TransMorph achieves the highest Dice and the lowest non-positive Jacobian ratio in the zero-shot group (roughly two orders of magnitude below plain TransMorph zero-shot), confirming that the folding-suppression effect transfers robustly across cohort and scanner (Section 3.5; Table 3; Figure 3).

From a clinical neuroimaging perspective, this gain directly addresses a long-standing usability barrier of learning-based registration: Jacobian-based biomarkers, atlas-fusion segmentation, and longitudinal morphometry pipelines all assume one-to-one, fold-free warps, an assumption that unconstrained Transformer backbones routinely violate. By delivering near-diffeomorphic fields at feed-forward inference speed and with no change to the deployment architecture, HypEReg-TransMorph offers a drop-in upgrade path for brain-MRI research and clinical-research workflows—spanning atrophy quantification in neurodegenerative disease, atlas-based delineation for radiotherapy and surgical planning, and population-scale morphometry—without sacrificing the throughput and reproducibility advantages that motivated the move from classical iterative methods to learned registration in the first place. The released open-source package, together with the remaining roadmap (multi-seed retraining and additional external cohorts), positions non-

linear Jacobian-aware regularization as a practical, low-cost building block for clinically deployable Transformer-based deformable registration.

Supplementary Materials: Supplementary Materials (separate PDF) include: Table S1 (inferential summary for selected baselines), Table S2 (paired inferential statistics with Wilcoxon+BH-FDR), Table S3 (grouped-VOI label mapping), Figure S1 (descriptive IXI multi-metric overview), OASIS training/environment details, Table S4 (term-by-term ablation per-case hooks), Table S5 (OASIS paired Wilcoxon/BH-FDR), implementation and profiling details, and additional qualitative registration/Jacobian visualizations.

Author Contributions: Conceptualization, S.X. and M.X.; methodology, S.X. and M.X.; software, S.X.; validation, S.X. and M.X.; formal analysis, S.X.; investigation, S.X.; data curation, S.X.; writing—original draft preparation, S.X.; writing—review and editing, S.X. and M.X.; visualization, S.X.; supervision, M.X.; project administration, M.X. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: Not applicable. This study performed secondary analysis of publicly available anonymized IXI data and did not recruit new human participants.

Informed Consent Statement: Not applicable to this secondary-analysis study using publicly available anonymized data.

Data Availability Statement: Source code, configuration files, per-case metric tables, and figure/statistics scripts are openly available at https://github.com/Zzxmh/HypEReg-TransMorph and archived at Zenodo: https://doi.org/10.5281/zenodo.19888526 [32]. Trained model weights are available from the corresponding author upon reasonable request. The IXI dataset used in this study is publicly available at https://brain-development.org/ixi-dataset/ (CC BY-SA 3.0). The preprocessed split follows the public TransMorph-style IXI preprocessing release.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

HypEReg	Hyperelastic-Energy Regularization
MRI	Magnetic Resonance Imaging
HD95	95th percentile Hausdorff Distance
ASSD	Average Symmetric Surface Distance
NCC	Normalized Cross-Correlation
LNCC	Local Normalized Cross-Correlation
SDlogJ	Standard Deviation of Log Jacobian Determinant
IXI	Information eXtraction from Images dataset
BMR	Burger–Modersitzki–Ruthotto (hyperelastic energy framework)
FDR	False Discovery Rate

References

1. Balakrishnan, G.; Zhao, A.; Sabuncu, M.R.; Guttag, J.; Dalca, A.V. VoxelMorph: A learning framework for deformable medical image registration. *IEEE Transactions on Medical Imaging* **2019**, *38*, 1788–1800. https://doi.org/10.1109/TMI.2019.2897538.
2. Kim, B.; Kim, D.H.; Park, S.H.; Kim, J.; Lee, J.G.; Ye, J.C. CycleMorph: Cycle consistent unsupervised deformable image registration. *Medical Image Analysis* **2021**, *71*, 102036. https://doi.org/10.1016/j.media.2021.102036.
3. Chen, J.; Frey, E.C.; He, Y.; Segars, W.P.; Li, Y.; Du, Y. TransMorph: Transformer for unsupervised medical image registration. *Medical Image Analysis* **2022**, *82*, 102615. https://doi.org/10.1016/j.media.2022.102615.

4. Burger, M.; Modersitzki, J.; Ruthotto, L. A Hyperelastic Regularization Energy for Image Registration. *SIAM Journal on Scientific Computing* **2013**, *35*, B132–B148. https://doi.org/10.1137/110835955.
5. Zou, J.; Debroux, N.; Liu, L.; Qin, J.; Schönlieb, C.B.; Aviles-Rivero, A.I. Conformal-Invariant Hyperelastic Regularisation for Deep Learning–Based Image Registration. In Proceedings of the Medical Image Computing and Computer Assisted Intervention (MICCAI), 2023. Hyperelastic regularization for deep registration; lung CT, https://doi.org/10.1007/978-3-031-43999-5_60.
6. Hering, A.; Häger, S.; Moltz, J.H.; Lessmann, N.; Heldmann, S.; van Ginneken, B. CNN-Based Lung CT Registration with Multiple Anatomical Constraints. *Medical Image Analysis* **2021**, *72*, 102139. Volume-change penalty plus curvature regularizer in CNN registration, https://doi.org/10.1016/j.media.2021.102139.
7. Mok, T.C.W.; Chung, A.C.S. Fast Symmetric Diffeomorphic Image Registration with Convolutional Neural Networks. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 4644–4653. Selective Jacobian determinant penalty on det $J \leq 0$ regions, https://doi.org/10.1109/CVPR42600.2020.00470.
8. Avants, B.B.; Epstein, C.L.; Grossman, M.; Gee, J.C. Symmetric diffeomorphic image registration with cross-correlation: evaluating automated labeling of elderly and neurodegenerative brain. *Medical Image Analysis* **2008**, *12*, 26–41. https://doi.org/10.1016/j.media.2007.06.004.
9. Rueckert, D.; Sonoda, L.I.; Hayes, C.; Hill, D.L.G.; Leach, M.O.; Hawkes, D.J. Nonrigid Registration Using Free-Form Deformations: Application to Breast MR Images. *IEEE Transactions on Medical Imaging* **1999**, *18*, 712–721. https://doi.org/10.1109/42.796284.
10. Ashburner, J.; Friston, K.J. Voxel-Based Morphometry—The Methods. *NeuroImage* **2000**, *11*, 805–821. https://doi.org/10.1006/nimg.2000.0582.
11. Dalca, A.V.; Balakrishnan, G.; Guttag, J.; Sabuncu, M.R. Unsupervised Learning of Probabilistic Diffeomorphic Registration for Images and Surfaces. *Medical Image Analysis* **2019**, *57*, 226–236. https://doi.org/10.1016/j.media.2019.07.006.
12. Liu, Z.; Lin, Y.; Cao, Y.; Hu, H.; Wei, Y.; Zhang, Z.; Lin, S.; Guo, B. Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. In Proceedings of the Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022. https://doi.org/10.1109/ICCV48922.2021.00986.
13. Xie, Y.; Zhang, J.; Shen, C.; Xia, Y. CoTr: Efficiently Bridging CNN and Transformer for 3D Medical Image Segmentation. In Proceedings of the Medical Image Computing and Computer Assisted Intervention (MICCAI), 2021, Vol. 12903, *Lecture Notes in Computer Science*, pp. 171–180. https://doi.org/10.1007/978-3-030-87199-4_16.
14. Zhou, H.Y.; Guo, J.; Zhang, Y.; Han, X.; Yu, L.; Wang, L.; Yu, Y. nnFormer: Volumetric Medical Image Segmentation via a 3D Transformer. *IEEE Transactions on Image Processing* **2023**, *32*, 4036–4045. https://doi.org/10.1109/TIP.2023.3293771.
15. Wang, W.; Xie, E.; Li, X.; Fan, D.P.; Song, K.; Liang, D.; Lu, T.; Luo, P.; Shao, L. PVT v2: Improved Baselines with Pyramid Vision Transformer. *Computational Visual Media* **2022**, *8*, 415–424. https://doi.org/10.1007/s41095-022-0274-8.
16. IXI Consortium. IXI Dataset. https://brain-development.org/ixi-dataset/, 2013. Available online: https://brain-development.org/ixi-dataset/ (accessed on 25 April 2026).
17. Kingma, D.P.; Ba, J. Adam: A Method for Stochastic Optimization. *International Conference on Learning Representations* **2015**. arXiv preprint, arXiv:1412.6980, https://doi.org/10.48550/arXiv.1412.6980.
18. Reddi, S.J.; Kale, S.; Kumar, S. On the Convergence of Adam and Beyond. In Proceedings of the International Conference on Learning Representations, 2018. OpenReview: https://openreview.net/forum?id=ryQu7f-RZ.
19. Avants, B.B.; Tustison, N.J.; Johnson, H.J. ANTsPyX: Advanced Normalization Tools in Python. https://github.com/ANTsX/ANTsPy, 2024. Software library; version 0.6.3 used in experiments.
20. Lowekamp, B.C.; Chen, D.T.; Ibáñez, L.; Blezek, D. The Design of SimpleITK. *Frontiers in Neuroinformatics* **2013**, *7*, 45. Software library; version 2.5.3 used in experiments, https://doi.org/10.3389/fninf.2013.00045.
21. Dice, L.R. Measures of the Amount of Ecologic Association Between Species. *Ecology* **1945**, *26*, 297–302. https://doi.org/10.2307/1932409.
22. Heimann, T.; van Ginneken, B.; Styner, M.A.; Arzhaeva, Y.; Aurich, V.; Bauer, C.; Beck, A.; Becker, C.; Beichel, R.; Bekes, G.; et al. Comparison and Evaluation of Methods for Liver Segmentation from CT Datasets. *IEEE Transactions on Medical Imaging* **2009**, *28*, 1251–1265. https://doi.org/10.1109/TMI.2009.2013851.
23. Hering, A.; Hansen, L.; Mok, T.C.W.; Chung, A.C.S.; Siebert, H.; Häger, S.; Lange, A.; Kuckertz, S.; Heldmann, S.; Shao, W.; et al. Learn2Reg: Comprehensive Multi-Task Medical Image Registration Challenge, Dataset and Evaluation in the Era of Deep Learning. *IEEE Transactions on Medical Imaging* **2023**, *42*, 697–712. https://doi.org/10.1109/TMI.2022.3213983.
24. Bookstein, F.L. Principal Warps: Thin-Plate Splines and the Decomposition of Deformations. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **1989**, *11*, 567–585. https://doi.org/10.1109/34.24792.
25. Mok, T.C.W.; Chung, A.C.S. Large Deformation Diffeomorphic Image Registration with Laplacian Pyramid Networks. In Proceedings of the Medical Image Computing and Computer Assisted Intervention, 2020, pp. 211–221. https://doi.org/10.1007/978-3-030-59713-9_21.

26. Studholme, C.; Hill, D.L.G.; Hawkes, D.J. An Overlap Invariant Entropy Measure of 3D Medical Image Alignment. *Pattern Recognition* **1999**, *32*, 71–86. https://doi.org/10.1016/S0031-3203(98)00091-0.
27. Wang, Z.; Bovik, A.C.; Sheikh, H.R.; Simoncelli, E.P. Image Quality Assessment: From Error Visibility to Structural Similarity. *IEEE Transactions on Image Processing* **2004**, *13*, 600–612. https://doi.org/10.1109/TIP.2003.819861.
28. Wilcoxon, F. Individual Comparisons by Ranking Methods. *Biometrics Bulletin* **1945**, *1*, 80–83. https://doi.org/10.2307/3001968.
29. Benjamini, Y.; Hochberg, Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B* **1995**, *57*, 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.
30. Greer, H.; Kwitt, R.; Vialard, F.X.; Niethammer, M. ICON: Learning Regular Maps Through Inverse Consistency. In Proceedings of the Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2021, pp. 3396–3405. https://doi.org/10.1109/ICCV48922.2021.00338.
31. Tian, L.; Greer, H.; Vialard, F.X.; Kwitt, R.; Estépar, R.S.J.; Rushmore, R.J.; Makris, N.; Bouix, S.; Niethammer, M. GradICON: Approximate Diffeomorphisms via Gradient Inverse Consistency. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023, pp. 18084–18094. https://doi.org/10.1109/CVPR52729.2023.01734.
32. Xu, S.; Xu, M. HypEReg-TransMorph: A Hyperelastic-Regularized Transformer Framework for Folding-Suppressed Deformable Brain MRI Registration. https://doi.org/10.5281/zenodo.19888526, 2026. Software archive; version history available via Zenodo Concept DOI, https://doi.org/10.5281/zenodo.19888526.

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